



IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

AICTE - EDUSKILLS VIRTUAL INTERNSHIP

Java Full Stack Development With Project

An 8-Week Professional Internship Presentation Report

INTERNSHIP DURATION

June 2026 – August 2026 (8 Weeks)

STUDENT CREDENTIALS

Shashwat Upadhyay

ENROLLMENT / ROLL NO.

2410030445

PROGRAM & SEMESTER

B.Tech (CSE)

HOST ORGANIZATION

EduSkills Academy

Supported by AICTE National Internship Portal

Candidate & Program Overview



Candidate Details

Student Name: Shashwat Upadhyay

Roll Number: 2410030445

Degree: B.Tech (Computer Science & Engineering)

Academic Batch: 2026-27

University: IILM University, Greater Noida, U.P.



Internship Profile

Program Name: AICTE – EduSkills Virtual Internship

Domain: Java Full Stack Development With Project

Duration: 8 Weeks (June 2026 – August 2026)

Certificate ID: 2026-B2D5388B58

AICTE Student ID: STU699786ca2a9381771538122

Organization Profile: EduSkills Academy

EduSkills Academy is an EdTech and skill-development organization that delivers the AICTE–EduSkills Virtual Internship Program, operating under the vision “*To empower students with industry-relevant skills.*”



AICTE Collaboration: This internship was conducted under the AICTE – EduSkills Virtual Internship Program, supported by the AICTE National Internship Portal.



Program Delivery: Training and project work were delivered fully online through the EduSkills Learning Management System (LMS).



Focus Area: Full-stack Java web development (HTML, CSS, Bootstrap, JavaScript, Spring Boot, Hibernate, MySQL), combined with applied, project-based learning.

Problem Statement & Academic Context



The Academic Gap

Most academic exercises expose students to search functionality only through ready-made libraries or search engines, leaving the underlying mechanics of text indexing and retrieval largely abstracted away.



The Project Scope

The Text Search Indexer was built from first principles as a Java console application that replicates core search-engine functionality — covering text normalization, tokenization, unique-word indexing, keyword matching and term-frequency counting.

Core Project Objectives



Text Normalization

Convert input text to lowercase and remove punctuation for consistent processing.



Tokenize & Index

Split text into tokens and build a unique-word index using a TreeSet.



Keyword Matching

Check a line of text for the presence of a specified search keyword.



Term Frequency Count

Quantify the occurrence of each word to support search-result ranking.

Technologies & Toolchain



Simulator Core

Language & Runtime:

Java, Scanner (System.in)



Text Processing

Techniques:

String methods, Regex (replaceAll)



Data Structures

Collections:

HashMap, TreeSet, ArrayList



Full-Stack Exposure

Frontend:

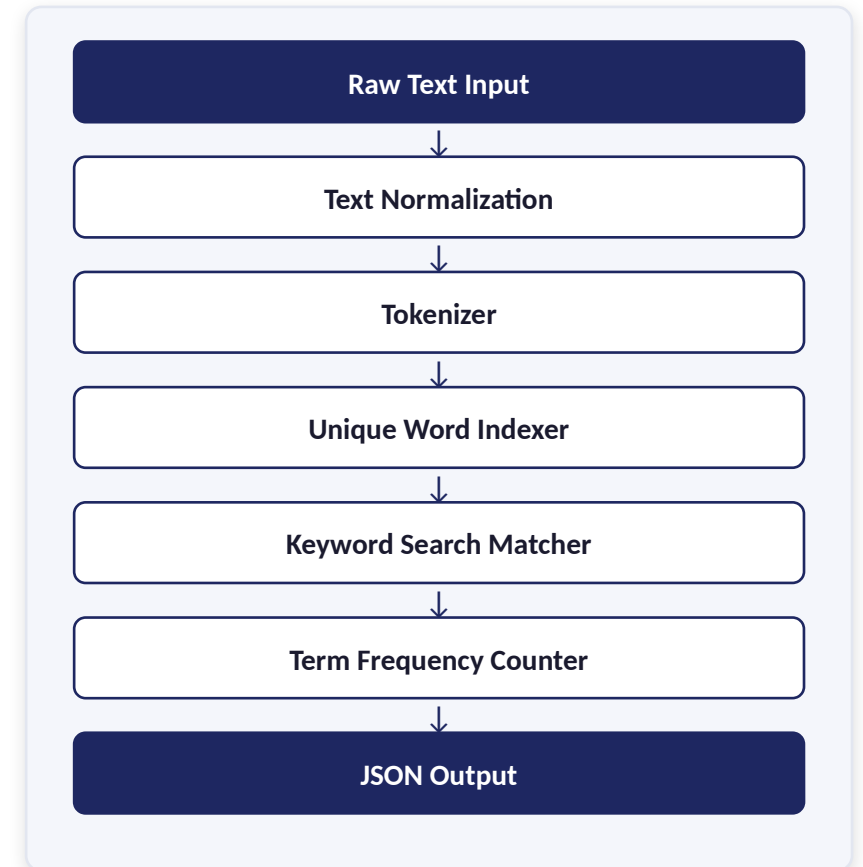
HTML, CSS, Bootstrap, JS/jQuery

Backend:

Spring Boot, Hibernate, MySQL, Git

System Architecture

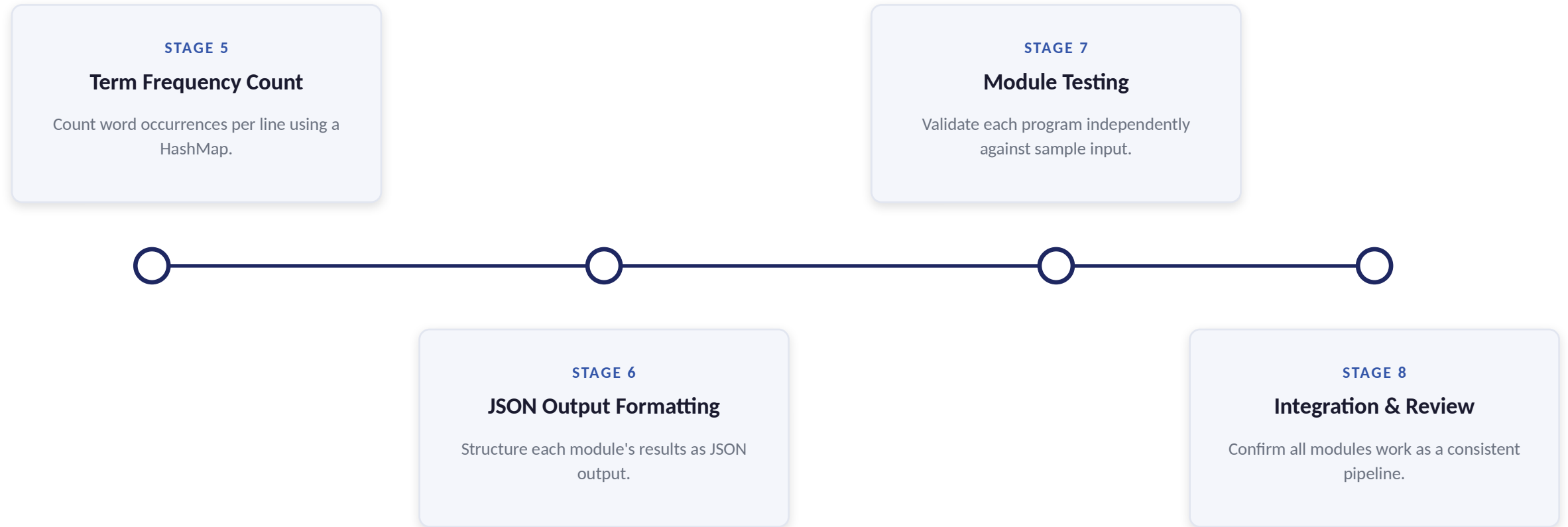
- 1 **Input Layer:** Raw text lines are supplied to the simulator via standard input (Scanner).
- 2 **Application Layer:** The Java console application receives input and coordinates all module operations.
- 3 **Data Layer:** Tokens and vocabulary are organised using Java Collections (HashMap, TreeSet).
- 4 **Operation Layer:** Normalization, tokenization, indexing, matching and frequency-counting logic executes.
- 5 **Output Layer:** Results for each processed line are serialised and returned as structured JSON.



Development Timeline (Stages 1-4)



Development Timeline (Stages 5–8)



Key Features & Demonstrated Skills

- ✓ **Text Normalization:** Implemented regex-based cleanup for consistent casing and punctuation handling.
- ✓ **Tokenization & Indexing:** Built a tokenizer and unique-word indexer using core Java collections (TreeSet).
- ✓ **Keyword Search Matching:** Implemented boolean keyword-presence checks across text lines.
- ✓ **Term Frequency Counting:** Used HashMap for $O(1)$ average-time word-count updates and lookups.
- ✓ **Full-Stack Exposure:** Gained hands-on exposure to HTML, CSS, Bootstrap, JavaScript/jQuery, Spring Boot, Hibernate and MySQL through the broader internship curriculum.

Certification & Official Verification

Verification Metric	Official Record / Credential Details
Internship Title	Java Full Stack Development With Project
Issuing Organization	EduSkills Academy
Program Support	AICTE National Internship Portal
Certificate ID	2026-B2D5388B58
Certificate Issued On	August 13, 2026
AICTE Student ID	STU699786ca2a9381771538122
APAAR ID	509739474904

Certification & Official Verification



Thank You!



Shashwat Upadhyay

Roll No: 2410030445



**School of Computer Science &
Engineering**

IILM University, Greater Noida, U.P.